

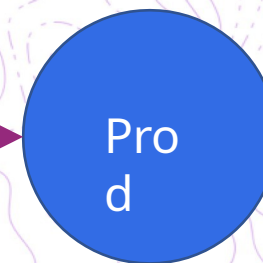
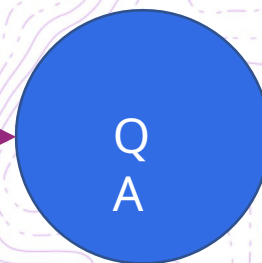
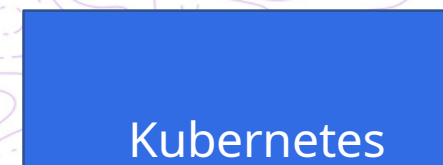
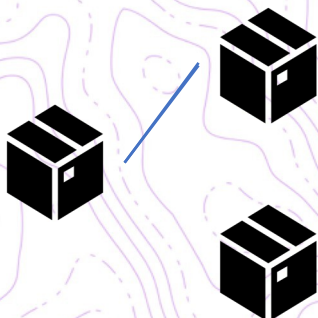
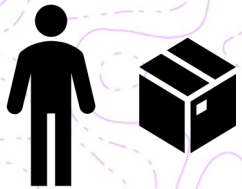


SAMSUNG SDS

Helm Is The Package Manager For Kubernetes

Helm은 Kubernetes의 패키지 관리자입니다

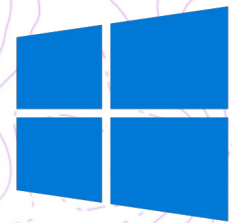




You Can Download And Run Helm For These Systems

다양한 OS 별 실행파일을 지원

macOS



Windows



A Similar Command Line Experience

예전과 유사한 커맨드 라인

```
$ helm repo add stable https://kubernetes-charts.storage.googleapis.com/
$ helm search repo mariadb
```

NAME	CHART VERSION	APP VERSION	DESCRIPTION
stable/mariadb	7.1.0	10.3.20	Fast, reliable, scalable, and easy to use open...
stable/phpmyadmin	4.2.4	4.9.2	phpMyAdmin is an mysql administration frontend

```
$ helm install mymaria stable/mariadb
$ helm list
```

NAME	NAMESPACE	REVISION	UPDATED	STATUS	CHART VERSION	APP	
mymaria	default	1	2019-11-27	-0500 EST	deployed	mariadb-7.1.0	10.3.20

```
$ helm uninstall mymaria
```


Namespaces!

네임 스페이스 !

```
$ helm list
```

NAME	NAMESPACE	REVISION	UPDATED	STATUS	CHART	APP VERSION
mymaria	default	1	2019-11-27 14:10:55.216169 -0500 EST	deployed	mariadb-7.1.0	10.3.20

```
$ helm ls --all-namespaces
```

NAME	NAMESPACE	REVISION	UPDATED	STATUS	CHART	APP VERSION
mymaria	default	1	2019-12-01 18:05:53.568566 -0500 EST	deployed	mariadb-7.1.0	10.3.20
mymaria2	foo	1	2019-12-01 18:06:12.520784 -0500 EST	deployed	mariadb-7.1.0	10.3.20

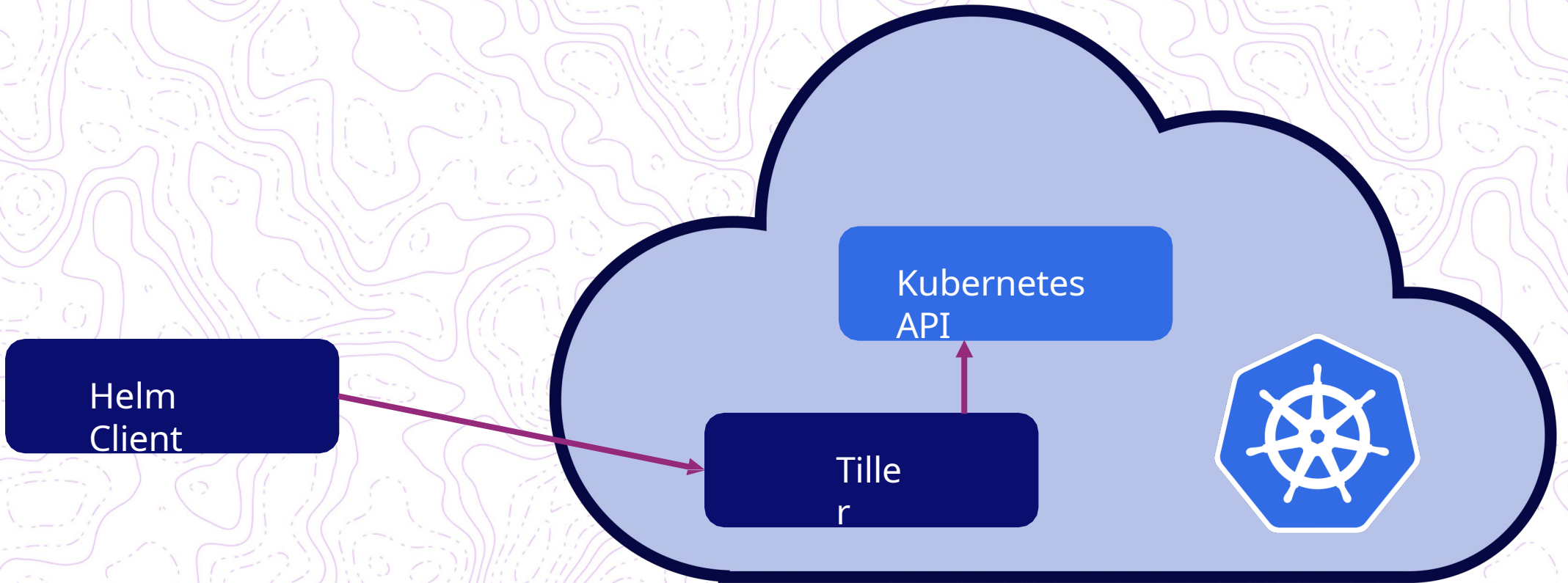
```
$ helm ls --namespace foo
```

NAME	NAMESPACE	REVISION	UPDATED	STATUS	CHART	APP VERSION
mymaria2	foo	1	2019-12-01 18:06:12.520784 -0500 EST	deployed	mariadb-7.1.0	10.3.20

```
$ helm install -n foo myNewMaria stable/mariadb
```

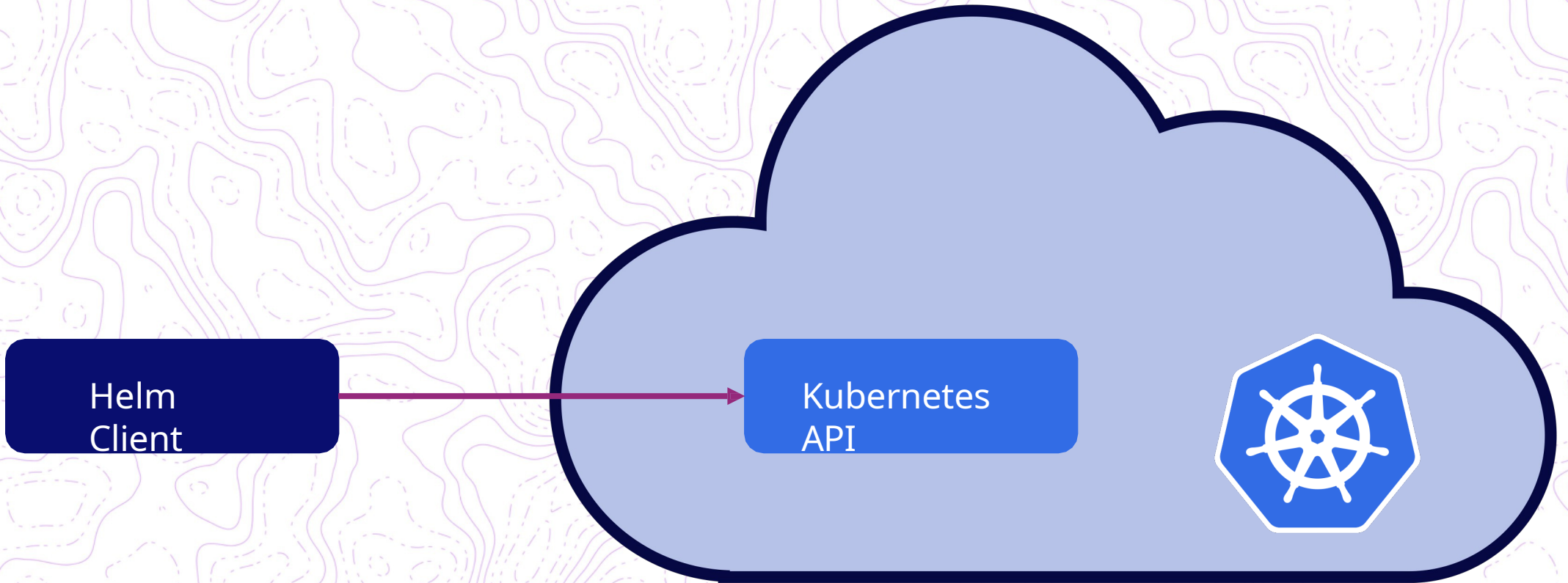
Helm v2 Architecture

Helm v2 아키텍처



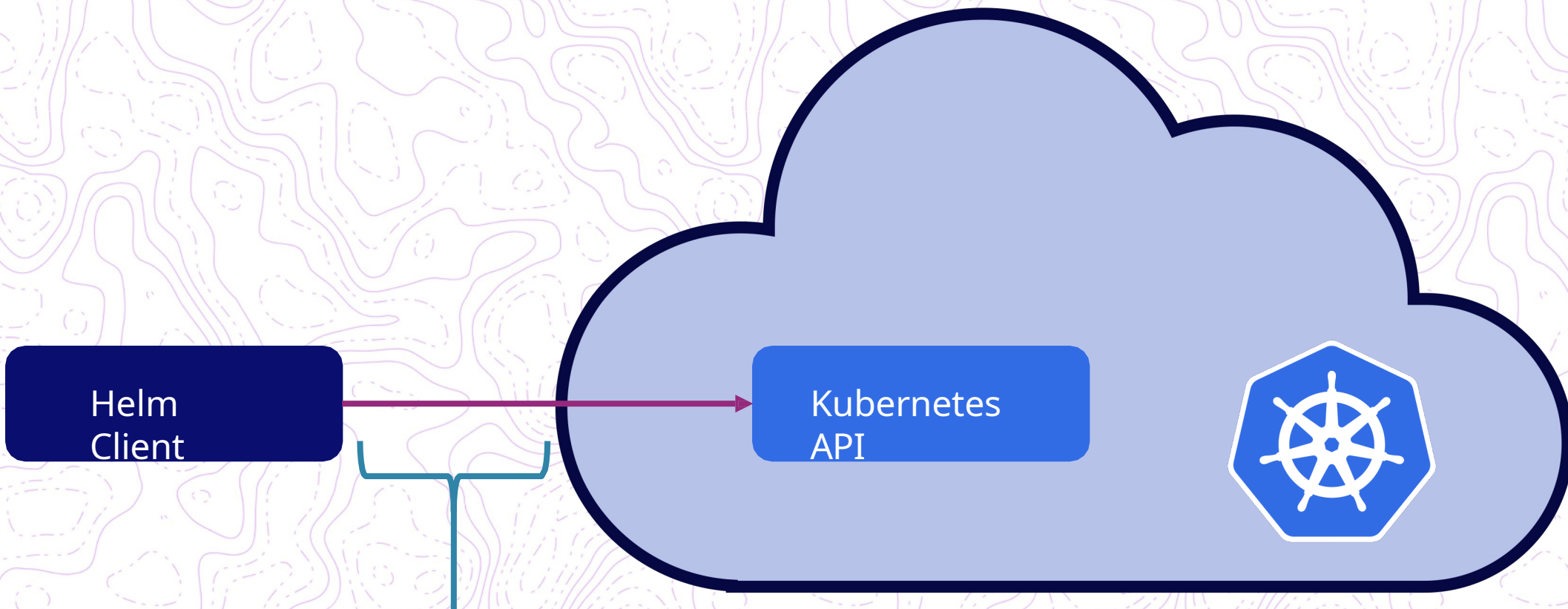
Helm v3 Architecture

Helm v3 아키텍처



Helm v3 Architecture

Helm v3 아키텍처

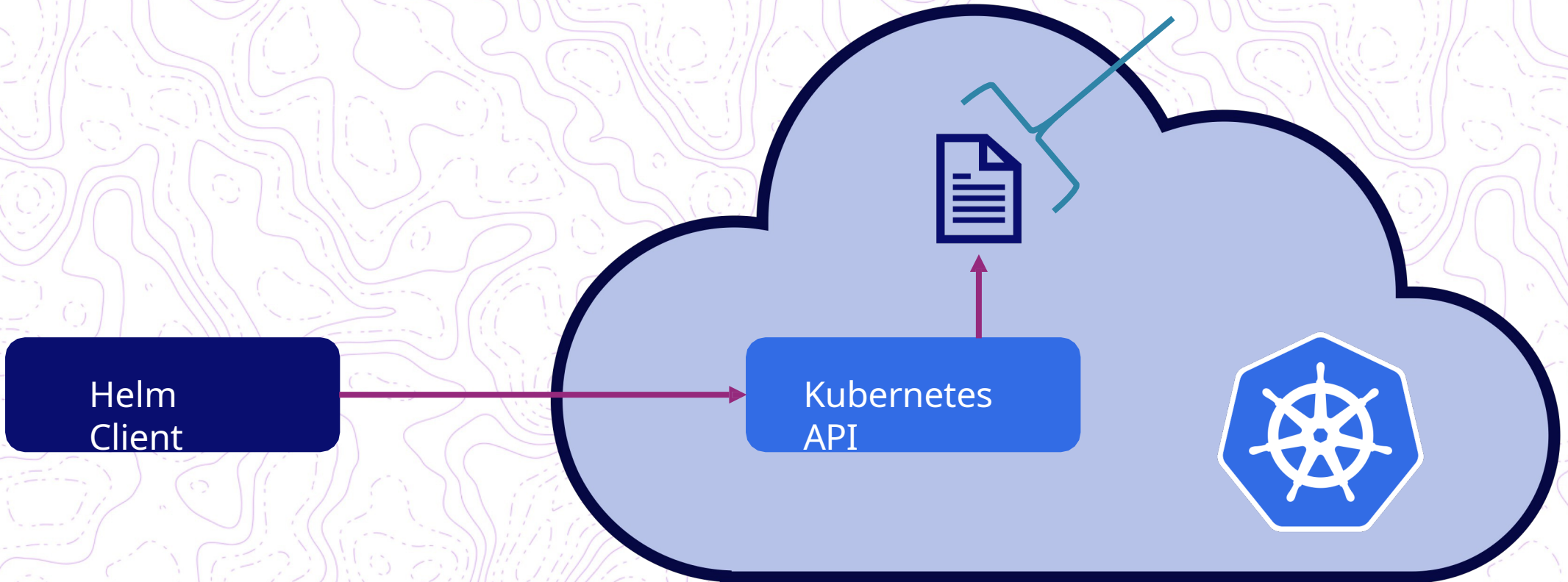


Uses normal user
credentials
일반 사용자 인증 사용

Helm v3 Architecture

Helm v3 아키텍처

Release information stored as ~~Secret~~ namespace with the app by default
앱이 설치된 네임스페이스에 릴리즈 정보가 *Secret* 으로 저장 (기본 설정)



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[Home](#)
[Introduction](#)
[How-To](#)[Topics](#)[Best Practices](#)
[Developing](#)[Templates](#)[Community](#)
[Frequently Asked Questions](#)[Glossary](#)[Contribute to Docs](#)You are viewing info for Helm 3 - check the [version FAQs](#) or [return to Helm 2](#) for prior versions.

Base Directory Support

The [XDG Base Directory Specification](#) is a portable standard defining where configuration, data, and cached files should be stored on the filesystem.

In Helm 2, Helm stored all this information in `~/.helm` (affectionately known as `helm home`), which could be changed by setting the `HELM_HOME` environment variable, or by using the global flag `--home`.

In Helm 3, Helm now respects the following environment variables as per the XDG Base Directory Specification:

- `$XDG_CACHE_HOME`
- `$XDG_CONFIG_HOME`
- `$XDG_DATA_HOME`

Helm plugins are still passed `$HELM_HOME` as an alias to `$XDG_DATA_HOME` for backwards compatibility with plugins looking to use `$HELM_HOME` as a scratchpad environment.

Several new environment variables are also passed in to the plugin's environment to accommodate this change:

- `$HELM_PATH_CACHE` for the cache path
- `$HELM_PATH_CONFIG` for the config path
- `$HELM_PATH_DATA` for the data path

Helm plugins looking to support Helm 3 should consider using these new environment variables instead.

directory helm.sh/he\m/v3/pkg v300

Published: Nov 12, 2019 | License: [Apache-2.0](#) | Module: [helm.sh/helm/v3](#)

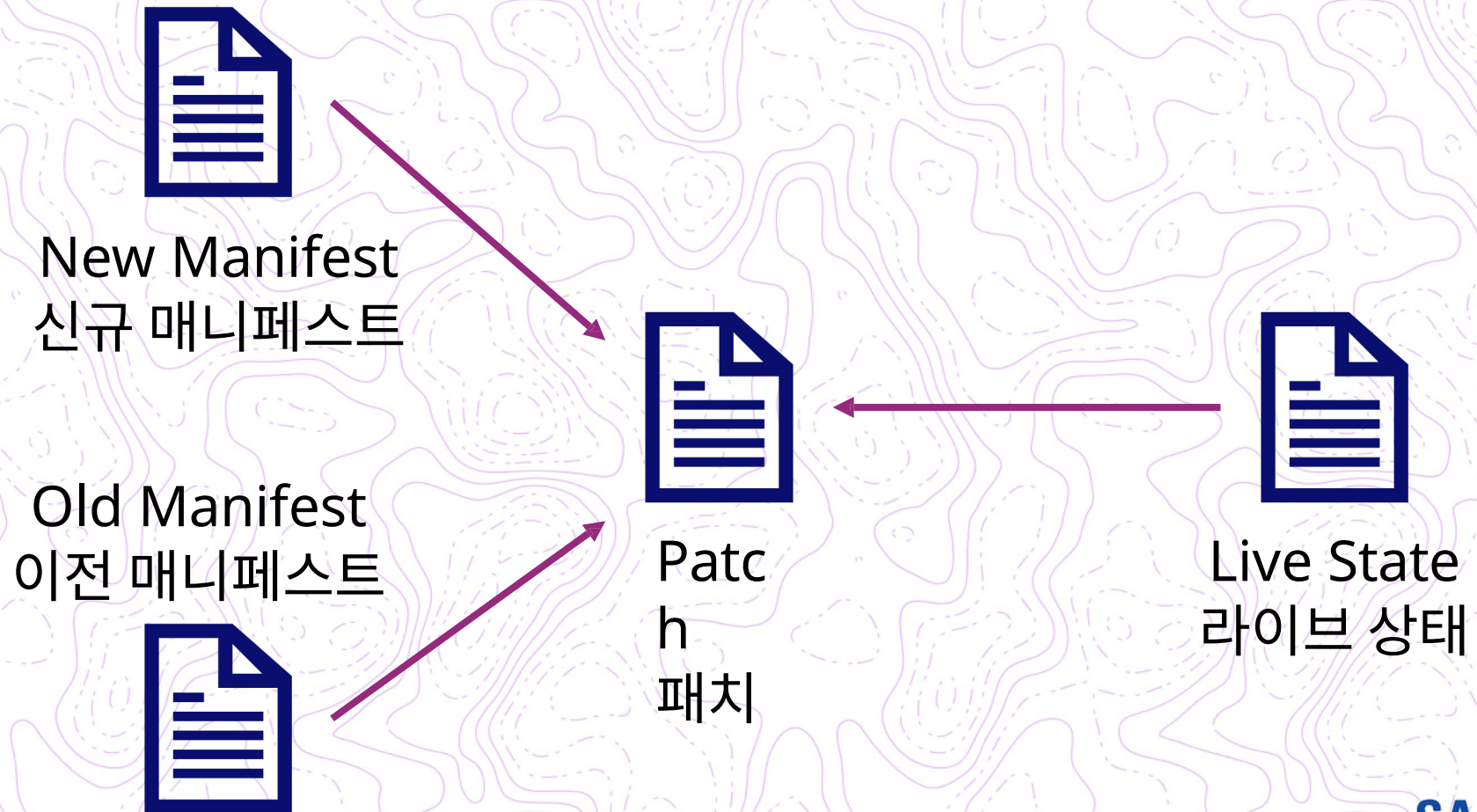
Doc
By

Overview Subdirectories Versions Imports Imported Licenses

Path	Synopsis
action	Package action contains the logic for each action that Helm can perform.
chart	
chart/loader	
chartutil	Package chartutil contains tools for working with charts.
cli	Package cli describes the operating environment for the Helm CLI.
cli/output	
cli/values	
downloader	Package downloader provides a library for downloading charts.
engine	Package engine implements the Go template engine as a Tiller Engine.
gates	Package gates provides a general tool for working with experimental feature gates.
getter	Package getter provides a generalize tool for fetching data by scheme.
helmpath	Unless required by applicable law or agreed to in writing, software distributed under the License is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
helmpath/xdg	
kube	
kube/fake	Package fake implements various fake KubeClients for use in testing

3-way Strategic Merge Patches

3 방향 전략적 병합 패치



3-way Strategic Merge Patches

3 방향 전략적 병합 패치

```
containers:  
- name: server  
  image: nginx:2.0.0
```



```
containers:  
- name: server  
  image: nginx:2.0.0  
- name: my-injected-  
  sidecar image: my-cool-  
  mesh:1.0.0
```


Additions To Charts

차트내 신규 기능 추가



```
$ tree newChart
newChart
├── Chart.yaml
├── charts
├── templates
├── NOTES.txt
├── helpers.tpl
├── deployment.yaml
├── ingress.yaml
├── service.yaml
├── serviceaccount.yaml
├── tests
├── test-connection.yaml
└── values.yaml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: {{ include "newChart.fullname" . }}
  labels:
    {{- include "newChart.labels" . |
      nindent 4 }}
spec:
  replicas: {{ .Values.replicaCount }}
  ...
```


apiVersion v2 For Helm 3

Helm 3 apiVersion v2

Chart.yaml

```
apiVersion: v2
name: mychart
description: A
Helm chart for
Kubernetes
```


Dependencies Now In Chart.yaml

현재 : Chart.yaml의 종속성

Chart.yaml

```
apiVersion: v2
name: mychart
description: A
Helm chart for
Kubernetes

dependencies:
- name:
  mariadb
  version:
  5.x.x
  repository: https://kubernetes-charts.storage.googleapis.com/
  condition: mariadb.enabled
  tags:
    - database
```


Library Charts

Chart.ya

ml

apiVersion: v2

name: mychart

description: A Helm chart for Kubernetes

type: library

JSON Schema Validation

JSON 스키마 유효성 검사

values.schema.js

```
on
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "properties": {
    "addresses": {
      "description": "List of addresses",
      "items": {
        "properties": {
          "city": {
            "type":
              "string"
          },
        },
      },
    },
  },
  ...
}
```


react-jsonschema-form

Simple

Nested

Arrays

Custom ArrayNumbers

Widgets

Single

Custom

Ordering

References

Custom

Errors

Large

Schema dependencies

Date & time

Validation

Files

Any Of

One Of

0 Live validation

default

Disable whole form

Omit extra data

Live omit

Object

JSONSchema

Alternatives

Null fields

Nullable

1 {

2 "title": "A registration form", Defaults

3 "description": "A simple form example.",

4 "type": "object",

5 "required": [

6 "firstName",

7 "lastName"

8],

9 "properties": {

10 "firstName": {

11 "type": "string",

12 "title": "First name",

13 "default": "Chuck"

14 },

15 "lastName": {

16 "type": "string",

17 "title": "Last name"

18 },

19 "age": {

20 "type": "integer",

21 "title": "Age"

22 },

}

UISchema

1 {

2 "firstName": {

3 "ui:autofocus": true,

4 "ui:emptyValue": ""

5 },

6 "age": {

7 "ui:widget": "updown",

8 "ui:title": "Age of person",

9 "ui:description": "(earthian year)"

10 },

11 "bio": {

formData

1 {

2 "firstName": "Chuck ",

3 "lastName": "Norris",

4 "age": 75,

5 "bio": "Roundhouse kicking since 1940",

6 "password": "noneed"

7 }

A registration form

A simple form example.

First name*

Chuck

Last name

Norris

Age of person

(earthian year)

75

Bio

Roundhouse kicking since 1940

Password

Hint: Make it strong!

Telephone

Submit

Share

Custom Resource Definitions (CRD)



```
$ tree -L 1 ambassador
```

```
ambassador
├── CHANGELOG.md
├── Chart.yaml
├── OWNERS
├── README.md
├── ci
├── crds
├── templates
└── values.yaml
```

```
$ tree ambassador/crds
```

```
ambassador/crds
├── authservice.yaml
├── consulresolver.yaml
├── filter.yaml
├── filterpolicy.yaml
├── kubernetesendpointresolver.yaml
├── kuberneteserviceresolver.yaml
├── mapping.yaml
├── module.yaml
├── ratelimit.yaml
├── ratelimitservice.yaml
├── tcpmapping.yaml
├── tlscontext.yaml
└── tracingservice.yaml
```


CRDs across Helm v2 and v3

Helm v2 및 v3 의 CRD

For Helm v3 Use The *crds*

Directory Helm v3 의 경우 crds
디렉토리를 사용하십시오 .

For Helm v2 Use A Template With *crd-install*

Hook Helm v2 의 경우 crd 설치 후크가있는
템플릿 사용

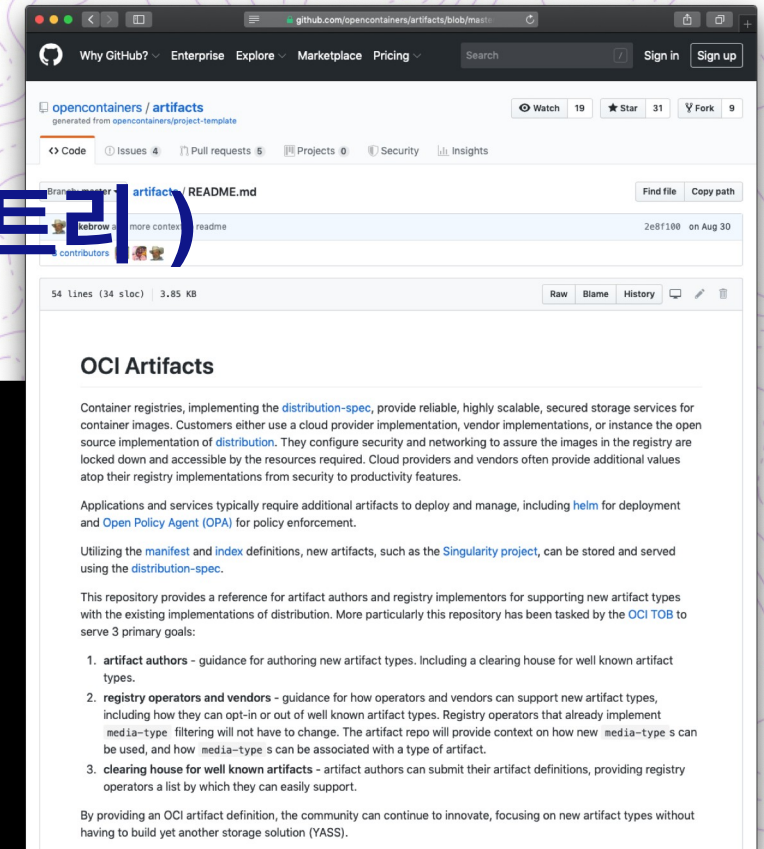
```
$ tree ambassador/crds
ambassador/crds
├── authservice.yaml
├── consulresolver.yaml
├── filter.yaml
├── filterpolicy.yaml
├── kubernetesendpointresolver.yaml
├── kubernetesresolver.yaml
├── mapping.yaml
├── module.yaml
├── ratelimit.yaml
├── ratelimitservice.yaml
└── tcpmapping.yaml
```

```
apiVersion: apiextensions.k8s.io/v1beta1
kind: CustomResourceDefinition
metadata:
  name: authservices.getambassador.io
  labels:
    app.kubernetes.io/name:
      ambassador
  annotations:
    "helm.sh/hook": crd-install
...
```

Helm Experiments (베타 기 능) OCI Registries (OCI 레지스트리)

```
$ export HELM_EXPERIMENTAL_OCI=1
$ docker run -dp 5000:5000 --restart=always --name registry registry
$ helm chart save mychart localhost:5000/myrepo/mychart:1.2.3
ref:      localhost:5000/myrepo/mychart:1.2.3
digest:   896935a875c8fe8f8b9b81e5862413de316f8da3d6d9a7e0f6f1e90f6204f551
size:     2.7 KiB
name:     mychart
version:  0.1.0
1.2.3:    saved
$ helm chart list
```

REF	NAME	VERSION	DIGEST	SIZE	CREATED
localhost:5000/myrepo/mychart:1.2.3	mychart	0.1.0	896935a	2.7 KiB	About a minute



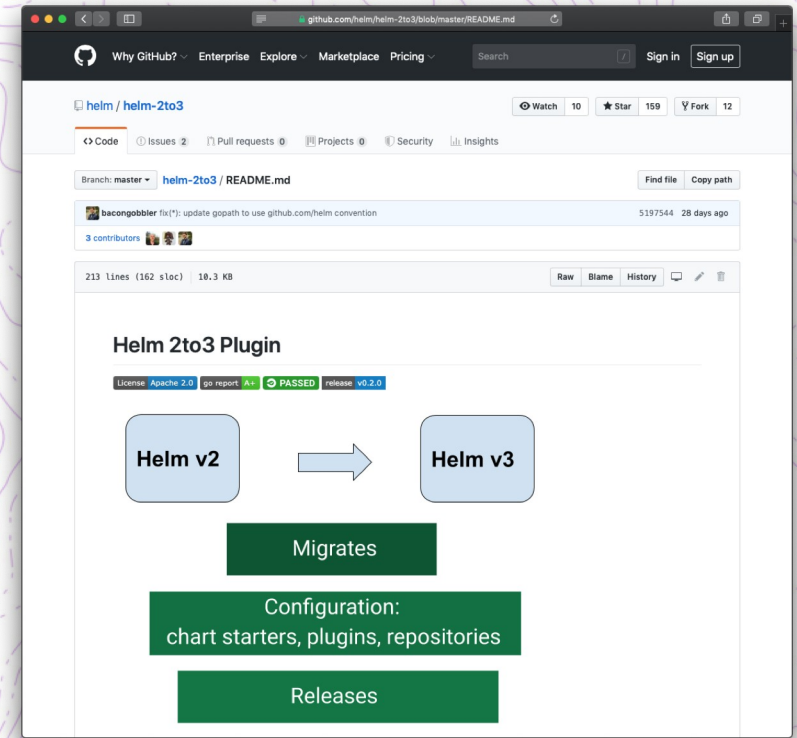
What about upgrading to Helm 3?

HelM 3 으로 업그레이드는 어떻게 할 수 있나요 ?

After all, the storage format changed

```
$ helm plugin install https://github.com/helm/helm-2to3
$ helm 2to3 move config
$ helm 2to3 convert RELEASE
$ helm 2to3 cleanup
```

* Read the docs to learn about flags like --dry-run
--dry-run 과 같은 플래그는 docs 를
확인해주세요





Learn more: <https://helm.sh>

Read Docs: <https://helm.sh/docs>

Source:

<https://github.com/helm/helm>

Mailing list: [cncf-](mailto:cncf-helm@lists.cncf.io)

helm@lists.cncf.io

<https://lists.cncf.io/g/cncf-helm>